CHAPTER 13 · DERIVATIONS

ELECTROSTATICS

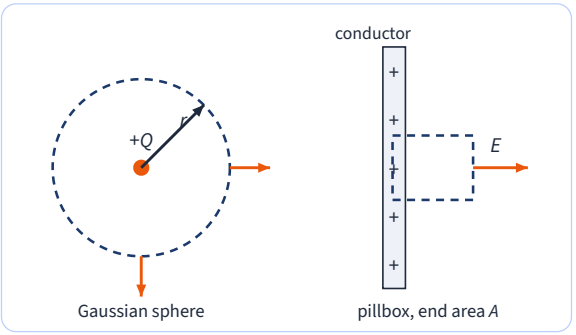
2 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 ELECTRIC FIELD STRENGTH FROM GAUSS'S LAW

SYLLABUS 13.3(D)

RESULT $E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$; sphere: 0 inside; plate: $E = \dfrac{\sigma}{\varepsilon_0}$

1. **Point charge Q**: Gaussian sphere of radius r centred on Q. E is radial and the same everywhere on it, so $\Phi = E(4\pi r^2) = \dfrac{Q}{\varepsilon_0}$, giving $\mathbf{E = \dfrac{Q}{4\pi\varepsilon_0 r^2}}$.
2. **Charged conducting sphere** (radius R, charge Q on its surface): for r > R the Gaussian sphere encloses Q, so $E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$; for r < R it encloses no charge, so **E = 0 inside**.
3. **Charged conducting plate** (surface charge density σ): a pillbox of end area A crosses the surface, one end inside the conductor (E = 0) and one end outside.
4. Flux only through the outer end: $EA = \dfrac{\sigma A}{\varepsilon_0}$, so $\mathbf{E = \dfrac{\sigma}{\varepsilon_0}}$, perpendicular to the surface.



2 GAUSS'S LAW IS EQUIVALENT TO COULOMB'S LAW

SYLLABUS 13.3

RESULT $F = Qq/(4\pi\varepsilon_0 r^2)$

1. Draw a Gaussian surface: a sphere of radius r centred on a point charge Q.
2. Apply Gauss's law: total flux through the closed surface $\Phi = \dfrac{Q}{\varepsilon_0}$.
3. Definition of electric flux: E is radial and the same at every point of the sphere, so $\Phi = E(4\pi r^2)$.
4. $E(4\pi r^2) = \dfrac{Q}{\varepsilon_0}$, so the electric field strength is $E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$.
5. Place a charge q at distance r and use the definition $E = \dfrac{F}{q}$: $F = qE$, so $\mathbf{F = Qq/(4\pi\varepsilon_0 r^2)}$, which is Coulomb's law.